Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/16/2018
PI Name:	Hollberg, Leo W Ph.D.		
Project Title:	Advancing Time Transfer and Optical Atomic Clocks for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stanford	**State:**	CA
Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	05/30/2012	**End Date:**	08/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per JPL info and PI (Ed., 9/22/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kahn, Joseph Ph.D. (Stanford University) Belmonte, Aniceto Ph.D. (University of Barcelona)		
Grant/Contract No.:	JPL 1458931		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our research project focuses on one of the most critical and significant scientific and technical challenges to using advanced cold-atom atomic clocks in space. The problems that we address are the serious performance limitations to the existing Time-Transfer links between ground and space. This research will advance both the basic science and the technology of Time-Transfer methods that could both support the current European Space Agency-Atomic Clock Ensemble in Space (ESA-ACES) mission plan and subsequently enhance performance in order to meet the demands of future optical atomic clocks (e.g., ESAs-SOC) for space. The improved performance is enabled by advanced laser methods for absolute Time and Frequency transfer that are based on stabilized femtosecond mode-locked lasers. We have developed a novel integrated system concept that will optimize the Time-Transfer performance in real-world systems by combining an ultrahigh performance Laser Time-Transfer (Laser-TT) link with an advanced microwave link and GPS signals. Analysis of the fundamental capabilities and limitations will guide the final system design, and a prototype Laser-TT link will be built in the laboratory and tested to critically evaluate the system performance. The demonstrations and testing of the Time-Transfer link will include two-way laser transmission via both free-space and optical fibers, and two-way microwave transmission via coaxial cable, which will give over-constrained redundant signals allowing us to identify systematic errors and uncertainties. Our Laser-TT link will improve the absolute timing precision by a factor of 100 times compared to the most advanced microwave links. Nonetheless, the highest performance, practical, and reliable systems operating in real-world conditions (with weather, turbulence, etc.) will require both microwave and optical channels. Critical analysis and testing of the most promising concepts will allow us to optimally integrate an advanced microwave link (as per ACES) with our higher performance Laser-TT link.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced precision for "Time" (epoch), time interval, and frequency transfer to space could be enabling for several science and technology missions and applications, including: International Space Station (ISS), ACES, tests of relativity, geodesy, gravity. Optical timing can improve the performance and precision of Global Satellite Navigation Systems (GNSS) such as GPS. It could also provide much improved precision and accuracy of precise orbit determination and through better timing and orbit could enhance capabilities and accuracy in geodesy. Potentially leverage capabilities with existing International Laser Ranging Systems to mutual benefit. Definitely strong synergism with satellite laser communication systems now under development.
Task Progress:	During this year we developed a theoretical model to understand the effects of atmospheric turbulence on the achievable precision of Time transfer and range measurements that could be achieved between ground and space using our two-way Doppler-cancelling laser link concepts. This analytical work is done in collaboration with Aniceto Belmonte of the University of Barcelona and Joseph Kahn in EE (Electrical Engineering). That work was published, and we will compare our predictions with experimental results. In the last several months of this project year we have been building and testing a free-space laser link between a roof top location on campus and the Student Observatory on the hills behind campus. The link covers a one way distance of 1 km. With that free-space link we have been measuring the Point Spread Function (PSF) of the atmospheric distortions and turbulence and also the phase front errors. Those measurements are done with a camera for the PSF and a Shack-Hartman wavefront sensor for the wavefront. Both of those detectors are coupled to the output of a small (3.5” aperture) good quality telescope. Currently we have a red laser (20 mW max) at the Student Observatory and the 3.5” telescope and the two sensors at ES-III. We also have optical corner cubes at both sites (rather small 1” and 2”) but reflection signals are visible. The system is producing valuable data on turbulence, S/N, wavefront, which serves for comparison of our theoretical models. We continue to explore methods and support opportunities that might enable a demonstration mission to the ISS. We believe that it is now feasible to put together space-qualified and qualifiable hardware that could connect high performance clocks on the ground with atomic clocks that would fly on the ISS. That system could consist of space lasercom terminals from other NASA programs or others, GPS Rb, an RF antenna to pick up radiated microwave signal from ACES-ESC clocks that will be on the ISS in the next few years. This would require a site on the ISS, a launch, a little power, some funding support to put everything together including mobile ground terminal based on commercial components; then we could have an optical link that could measure the ACES clocks relative to ground clocks in the US (NIST--National Institute of Standards & Technology, JPL--Jet Propulsion Laboratory).
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Articles in Peer-reviewed Journals	Belmonte A, Taylor MT, Hollberg L, Kahn JM. "Effect of atmospheric anisoplanatism on earth-to-satellite time transfer over laser communication links." Opt Express. 2017 Jul 10;25(14):15676-86. https://doi.org/10.1364/OE.25.015676 ; PubMed PMID: 28789081 , Jul-2017